

# CNG KSP Feature Matrix

Every capability observed across shipping CNG Key Storage Providers, with this project's coverage and — where a gap exists — what closing it would take. Generated 2026-09-21.

|                      |         |         |             |               |                 |
|----------------------|---------|---------|-------------|---------------|-----------------|
| 99                   | 52      | 9       | 38          | 53%           | 39              |
| Capabilities tracked | Covered | Partial | Not covered | Fully covered | Actionable gaps |

## How to read this

| Status      | Meaning                                                                                    | Ev | Evidence for the market claim                                                                              |
|-------------|--------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------|
| Covered     | Implemented and exercised by the test suite.                                               | A  | Read from source code or an authoritative API reference that enumerates the provider's own algorithm list. |
| Partial     | Implemented, but reachable only through a non-standard identifier or with a stated caveat. | B  | Stated in vendor or Microsoft documentation surfaced through search; primary page not read in full.        |
| Not covered | Absent. The Gap column says what closing it would take.                                    | C  | Drawn from HSM datasheets describing the device rather than its KSP. Treat as a lead, not a fact.          |

**Effort** on gaps: S = a few lines to a day · M = days · L = a substantial piece of work. **N/A** marks a deliberate position or an external blocker rather than a backlog item — those are excluded from the actionable-gap count.

| ID                 | Feature                               | Detail                                                          | Market support  | Ev | Status      | Gap — what closing it would take                                                                                                                                                                                                                                                                                                                                                                                                               | Eff |
|--------------------|---------------------------------------|-----------------------------------------------------------------|-----------------|----|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| PROVIDER INTERFACE |                                       |                                                                 |                 |    |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| IFACE-01           | GetKeyStorageInterface export         | Sole exported symbol; returns the KSP function table            | All providers   | A  | Covered     | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | —   |
| IFACE-02           | OpenProvider / FreeProvider           | Provider handle lifecycle                                       | All providers   | A  | Covered     | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | —   |
| IFACE-03           | GetProviderProperty                   | Name / Version / ImplType                                       | All providers   | A  | Covered     | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | —   |
| IFACE-04           | SetProviderProperty                   | Provider-level configuration                                    | Vendor-specific | C  | Not covered | Stubbed as NTE_NOT_SUPPORTED. Would need a defined config surface — slot selection and PIN source. Low value until multi-slot exists.                                                                                                                                                                                                                                                                                                          | M   |
| IFACE-05           | NotifyChangeKey                       | Key-change notification handle                                  | Microsoft KSP   | B  | Not covered | Returns ERROR_SUCCESS as a no-op. Real support needs a registry/file watcher on the token store and an event object per caller.                                                                                                                                                                                                                                                                                                                | L   |
| IFACE-06           | GetOperationProperty                  | Async / overlapped operation state                              | Microsoft KSP   | B  | Not covered | Returns NTE_NOT_SUPPORTED. Only needed for overlapped I/O; PKCS#11 is synchronous so there is nothing to report.                                                                                                                                                                                                                                                                                                                               | L   |
| IFACE-07           | PromptUser                            | Interactive PIN / consent dialog                                | Smart card KSPs | B  | Not covered | Returns NTE_NOT_SUPPORTED. Would need a UI thread and a window handle from NCrypt_■WINDOW_■HANDLE_■PROPERTY.                                                                                                                                                                                                                                                                                                                                   | L   |
| IFACE-08           | FreeBuffer / FreeObject               | Provider-allocated memory release                               | All providers   | A  | Covered     | —                                                                                                                                                                                                                                                                                                                                                                                                                                              | —   |
| TABLE-01           | Function table layout matches the SDK | NCrypt_■KEY_■STORAGE_■FUNCTION_■TABLE slot order and membership | All providers   | A  | Partial     | The table now uses designated initialisers, so the Windows SDK header assigns the slots by name and a wrong name is a build error rather than a displaced slot. IsAlgSupported, EnumAlgorithms and VerifySignature are implemented and wired; the invented GetOperationProperty and FreeObject slots are gone. test_■function_■table.c covers ksp_main.c for the first time. Remaining: confirm against the real ncrypt_provider.h on Windows. | S   |
| BUILD-01           | Builds for Windows at all             | The shipped artefact is a Windows DLL                           | All providers   | A  | Partial     | Phase 0: nine of the ten source files now cross-compile clean for Windows. ksp_main.c cannot be verified here because it needs the WDK ncrypt_provider.h, which mingw-w64 does not ship; the CI windows job is what settles it. All six root causes are fixed.                                                                                                                                                                                 | M   |

| ID               | Feature                                  | Detail                                          | Market support            | Ev | Status      | Gap — what closing it would take                                                                                                                                                                                                                                                                                        | Eff |
|------------------|------------------------------------------|-------------------------------------------------|---------------------------|----|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| BUILD-02         | Continuous integration                   | Automated build and test on the target platform | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ASYMMETRIC RSA   |                                          |                                                 |                           |    |             |                                                                                                                                                                                                                                                                                                                         |     |
| RSA-01           | RSA key generation                       | 2048 / 3072 / 4096 bits                         | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-02           | RSA 1024 and below                       | 512-1024 bits                                   | MS Software KSP; Utimaco  | B  | Partial     | Build-time opt-in only. KSP_RSA_MIN_BITS defaults to 2048; a legacy build can lower it, but the shipped default still rejects <2048 with NTE_BAD_LEN. Deliberate: silently allowing 1024 would weaken every deployment.                                                                                                 | S   |
| RSA-03           | RSA above 4096                           | 8192 / 16384 bits                               | MS Software KSP; Utimaco  | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-04           | RSA arbitrary size increments            | 64-bit (MS) or 8-bit (Utimaco) steps            | MS Software KSP; Utimaco  | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-05           | Custom public exponent                   | Exponent other than 65537                       | Vendor-specific           | C  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-06           | PKCS#1 v1.5 signing                      | SHA-1 / 256 / 384 / 512                         | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-07           | PSS signing                              | SHA-1 / 256 / 384 / 512 with caller salt length | All providers             | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-08           | PSS with SHA-224                         | SHA-224 salt and MGF1                           | Unknown                   | C  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-09           | PKCS#1 v1.5 decryption                   | AT_KEYEXCHANGE keys                             | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-10           | OAEP decryption                          | SHA-1 / 224 / 256 / 384 / 512                   | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-11           | OAEP application label                   | pbLabel / cbLabel passed through                | Unknown                   | C  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| RSA-12           | Raw RSA (no padding)                     | CKM_RSA_X_509                                   | Some HSMS                 | C  | Not covered | SoftHSM2 does not implement CKM_RSA_X_509. Blocked by the backend not by the KSP; would work against an HSM that offers it.                                                                                                                                                                                             | M   |
| ASYMMETRIC ECDSA |                                          |                                                 |                           |    |             |                                                                                                                                                                                                                                                                                                                         |     |
| ECDSA-01         | ECDSA P-256                              | secp256r1; 64-byte signature                    | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDSA-02         | ECDSA P-384                              | secp384r1; 96-byte signature                    | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDSA-03         | ECDSA P-521                              | secp521r1; 132-byte signature                   | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDSA-04         | Brainpool curves                         | brainpoolP256r1 / 384r1 / 512r1                 | Entrust; Utimaco (device) | C  | Not covered | Add the DER OIDs to config.h and P11_GetCurveOid. SoftHSM2 supports them. CNG has BCRYPT_ECC_CURVE_BRAINPOOL* names so this stays standards-compliant.                                                                                                                                                                  | M   |
| ECDSA-05         | secp256k1                                | Bitcoin curve                                   | Entrust (device)          | C  | Partial     | Curve, OID and mechanism are wired and the key generates and signs. Reachable only through this KSP's own ECDSA_SECP256K1 identifier; the standard CNG route is the generic ECDSA algorithm with BCRYPT_ECC_CURVE_NAME set to BCRYPT_ECC_CURVE_SECP256K1, which needs curve-name property handling in ksp_properties.c. | S   |
| ECDSA-06         | DER to r  s conversion                   | PKCS#11 returns DER; CNG wants raw              | All providers             | A  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| KEY AGREEMENT    |                                          |                                                 |                           |    |             |                                                                                                                                                                                                                                                                                                                         |     |
| ECDH-01          | ECDH P-256 / P-384 / P-521               | CKM_ECDH1_DERIVE with CKD_NULL                  | MS Software KSP; Utimaco  | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDH-02          | SecretAgreement / DeriveKey / FreeSecret | NCRYPT_SECRET_HANDLE lifecycle                  | MS Software KSP; Utimaco  | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDH-03          | BCRYPT_KDF_RAW_SECRET                    | Raw Z value returned                            | MS Software KSP           | B  | Covered     | —                                                                                                                                                                                                                                                                                                                       | —   |
| ECDH-04          | BCRYPT_KDF_HASH                          | SP 800-56A concatenation KDF                    | MS Software KSP           | B  | Not covered | Returns NTE_NOT_SUPPORTED. Implement by hashing the Z value with the caller-supplied algorithm and info from the NCryptBufferDesc. Moderate work; callers can run it with BCrypt instead.                                                                                                                               | M   |

| ID               | Feature                          | Detail                          | Market support                     | Ev | Status      | Gap — what closing it would take                                                                                                                                                                      | Eff |
|------------------|----------------------------------|---------------------------------|------------------------------------|----|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ECDH-05          | BCRYPT_KDF_HMAC / TLS_PRF / HKDF | Protocol-specific KDFs          | MS Software KSP                    | B  | Not covered | Same shape as ECDH-04. HKDF is the one worth doing first for TLS 1.3.                                                                                                                                 | M   |
| ECDH-06          | Finite-field Diffie-Hellman      | DH 512-4096                     | MS Software KSP                    | B  | Not covered | Out of scope by choice. ECDH supersedes it and no surveyed HSM KSP exposes it. Would need CKM_DH_PKCS_KEY_PAIR_GEN plus CKM_DH_PKCS_DERIVE.                                                           | L   |
| ASYMMETRIC EDDSA |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |
| EDDSA-01         | Ed25519 signing                  | 64-byte raw signature           | None                               | A  | Partial     | Works against SoftHSM2 but uses the non-standard identifier EDDSA_ED25519. CNG defines no EdDSA algorithm ID so no stock application can request it. Nothing to fix in the KSP; blocked on Microsoft. | N/A |
| EDDSA-02         | Ed448 signing                    | 114-byte raw signature          | None                               | A  | Partial     | Same constraint as EDDSA-01.                                                                                                                                                                          | N/A |
| EDDSA-03         | X25519 key agreement             | Curve25519 ECDH                 | MS CNG has BCRYPT_EC_C_CURVE_25519 | B  | Not covered | This one IS standard in CNG (Windows 10+) unlike Ed25519 signing. Add the curve OID and route it through the existing ECDH path. Highest-value gap in this table.                                     | M   |
| ASYMMETRIC DSA   |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |
| DSA-01           | DSA signing                      | 512-1024 bits                   | MS Software KSP                    | B  | Not covered | Deliberately out of scope. Deprecated in modern Windows PKI and absent from every HSM KSP surveyed.                                                                                                   | L   |
| POST-QUANTUM     |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |
| PQC-01           | ML-DSA (Dilithium)               | FIPS 204                        | Thales Luna (device claim)         | C  | Not covered | SoftHSM2 2.7.0 has no PQC mechanisms so this is blocked at the backend. CNG gained PQC identifiers in recent Windows builds; revisit when both sides support it.                                      | L   |
| PQC-02           | ML-KEM (Kyber)                   | FIPS 203                        | Thales Luna (device claim)         | C  | Not covered | Same blocker as PQC-01.                                                                                                                                                                               | L   |
| PQC-03           | LMS / XMSS                       | Stateful hash-based signatures  | Thales Luna (device claim)         | C  | Not covered | Same blocker as PQC-01. Stateful schemes also need key-state persistence the KSP has no model for.                                                                                                    | L   |
| SYMMETRIC AES    |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |
| AES-01           | AES key generation               | 128 / 192 / 256 bits            | None via KSP                       | A  | Covered     | —                                                                                                                                                                                                     | —   |
| AES-02           | AES-ECB                          | No IV                           | None via KSP                       | A  | Covered     | —                                                                                                                                                                                                     | —   |
| AES-03           | AES-CBC                          | 16-byte IV                      | None via KSP                       | A  | Covered     | —                                                                                                                                                                                                     | —   |
| AES-04           | AES-CBC with padding             | NCRYPT_PAD_CIPHER_FLAG_LAG      | None via KSP                       | A  | Covered     | —                                                                                                                                                                                                     | —   |
| AES-05           | AES-GCM                          | 12-byte nonce; 128-bit tag; AAD | None via KSP                       | A  | Covered     | —                                                                                                                                                                                                     | —   |
| AES-06           | AES-CTR                          | 16-byte counter block           | None via KSP                       | A  | Partial     | Works but ChainingModeCTR is a KSP extension. CNG defines no standard counter-mode string so no stock application will select it.                                                                     | N/A |
| AES-07           | AES-CCM                          | Authenticated mode              | None via KSP                       | A  | Not covered | Returns NTE_NOT_SUPPORTED. SoftHSM2 2.7.0 does not implement CKM_AES_CCM so this is a backend blocker not a KSP one.                                                                                  | M   |
| AES-08           | AES-CFB                          | Cipher feedback                 | None via KSP                       | A  | Not covered | Returns NTE_NOT_SUPPORTED. No SoftHSM2 mechanism; would need CKM_AES_CFB128 which 2.7.0 lacks.                                                                                                        | M   |
| AES-09           | AES key wrap                     | RFC 3394 / 5649                 | Some HSMs (device)                 | C  | Not covered | SoftHSM2 offers CKM_AES_KEY_WRAP and KEY_WRAP_PAD. Needs KSP_ExportKey / ImportKey wired to C_WrapKey / C_UnwrapKey plus a CNG blob convention.                                                       | L   |
| AES-10           | AES-CMAC                         | Block-cipher MAC                | Some HSMs (device)                 | C  | Not covered | SoftHSM2 offers CKM_AES_CMAC. Would route through KSP_SignHash like HMAC does; same non-standard-identifier caveat applies.                                                                           | M   |
| SYMMETRIC MAC    |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |
| HMAC-01          | HMAC-SHA1 / 256 / 384 / 512      | Generic secret keys             | None via KSP                       | A  | Partial     | Works against SoftHSM2 but HMAC_SHA* are our own identifiers. CNG does HMAC through BCrypt algorithm handles not KSP key algorithms.                                                                  | N/A |
| HMAC-02          | HMAC-SHA224                      | SHA-224 variant                 | None                               | C  | Covered     | —                                                                                                                                                                                                     | —   |
| SYMMETRIC LEGACY |                                  |                                 |                                    |    |             |                                                                                                                                                                                                       |     |

| ID             | Feature                         | Detail                                          | Market support           | Ev | Status      | Gap — what closing it would take                                                                                                                                     | Eff |
|----------------|---------------------------------|-------------------------------------------------|--------------------------|----|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 3DES-01        | 3DES / DES                      | 56 / 112 / 168 bits                             | None via KSP             | B  | Not covered | Deliberately out of scope. Deprecated and excluded from modern HLK requirements despite SoftHSM2 supporting 14 DES mechanisms.                                       | L   |
| KEY FORMATS    |                                 |                                                 |                          |    |             |                                                                                                                                                                      |     |
| FMT-01         | BCRYPT_RSAPUBLIC_BLOB export    | Header + exponent + modulus                     | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| FMT-02         | BCRYPT_ECCPUBLIC_BLOB export    | Header + X + Y                                  | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| FMT-03         | EdDSA public blob export        | Generic ECC magic + raw point                   | None                     | A  | Covered     | —                                                                                                                                                                    | —   |
| FMT-04         | BCRYPT_RSAPUBLIC_BLOB import    | Creates a session public key object             | Vendor-specific          | C  | Covered     | —                                                                                                                                                                    | —   |
| FMT-05         | BCRYPT_ECCPUBLIC_BLOB import    | Creates a session public key object             | Vendor-specific          | C  | Covered     | —                                                                                                                                                                    | —   |
| FMT-06         | Private key export              | RSAPUBLICPRIVATE / ECCPRIVATE                   | None (all refuse)        | A  | Not covered | Refused by design with NTE_NOT_SUPPORTED. Keys are CKA_■EXTRACTABLE=FALSE. This is correct HSM behaviour not a gap.                                                  | N/A |
| FMT-07         | Private key import              | Wrapping an external private key onto the token | Some HSMs                | C  | Not covered | Returns NTE_NOT_SUPPORTED. Would need C_CreateObject with private material or C_UnwrapKey. Contradicts the HSM posture; only add if a migration workflow demands it. | L   |
| FMT-08         | BCRYPT_KEY_DATA_BLOB            | Raw symmetric key material                      | MS Software KSP          | B  | Not covered | AES keys are created CKA_■EXTRACTABLE=FALSE so export cannot work. Import would need C_CreateObject with CKA_VALUE. Constants are already in the mock header.        | M   |
| FMT-09         | PKCS#8 blob                     | NCRYPT_■PKCS8_■PRIVATE_■KEY_■BLOB               | MS Software KSP          | B  | Not covered | Same constraint as FMT-07: it is a private-key format and keys are non-extractable.                                                                                  | L   |
| FMT-10         | OPAQUETRANSFER blob             | Vendor-opaque key transport                     | Vendor-specific          | C  | Not covered | Would let keys move between instances of the same provider. Needs a defined wrapping format; no standard to follow.                                                  | L   |
| KEY PROPERTIES |                                 |                                                 |                          |    |             |                                                                                                                                                                      |     |
| PROP-01        | NCRYPT_ALGORITHM_PROPERTY       | Read                                            | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-02        | NCRYPT_LENGTH_PROPERTY          | Read and write before finalize                  | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-03        | NCRYPT_KEY_TYPE_PROPERTY        | AT_SIGNATURE / AT_KEYEXCHANGE                   | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-04        | NCRYPT_NAME / UNIQUE_NAME       | Key label                                       | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-05        | NCRYPT_EXPORT_POLICY_PROPERTY   | Always 0 (non-exportable)                       | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-06        | NCRYPT_KEY_USAGE_PROPERTY       | Signing / decrypt / key agreement flags         | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-07        | NCRYPT_ALGORITHM_GROUP_PROPERTY | RSA / ECDSA / ECDH / EDDSA / AES / HMAC         | All providers            | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-08        | NCRYPT_CHAINING_MODE_PROPERTY   | Read and write on symmetric keys                | MS Software KSP (BCrypt) | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-09        | NCRYPT_INITIALIZATION_VECTOR    | Read and write on symmetric keys                | MS Software KSP (BCrypt) | A  | Covered     | —                                                                                                                                                                    | —   |
| PROP-10        | NCRYPT_BLOCK_LENGTH_PROPERTY    | AES block size                                  | MS Software KSP          | B  | Covered     | —                                                                                                                                                                    | —   |

| ID                   | Feature                                  | Detail                               | Market support                | Ev | Status      | Gap — what closing it would take                                                                                                                                          | Eff |
|----------------------|------------------------------------------|--------------------------------------|-------------------------------|----|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| PROP-11              | <b>NCRYPT_SECURITY_DESCR_PROPERTY</b>    | Per-key ACL                          | MS Software KSP               | B  | Not covered | Returns NTE_NOT_SUPPORTED. PKCS#11 has no ACL model; would need a side-channel store mapping key labels to SDDL strings.                                                  | L   |
| PROP-12              | <b>NCRYPT_UI_POLICY_PROMPT</b>           | Strong-key protection prompt         | MS Software KSP; smart cards  | B  | Not covered | Depends on PromptUser (IFACE-07). Same UI-thread requirement.                                                                                                             | L   |
| PROP-13              | <b>NCRYPT_PIN_PROPERTY</b>               | Per-key PIN                          | Smart card KSPs               | B  | Not covered | PIN is process-wide from SOFTHSM2_PIN. Per-key PINs would need a per-handle credential cache and re-login per operation.                                                  | M   |
| PROP-14              | <b>NCRYPT_CERTIFICATE_PROPERTY</b>       | Certificate stored alongside the key | Smart card KSPs               | B  | Not covered | Would store a CKO_CERTIFICATE object with the same CKA_LABEL. SoftHSM2 supports certificate objects so this is straightforward.                                           | M   |
| PROP-15              | <b>NCRYPT_WINDOW_HANDLE_PROPERTY</b>     | Parent HWND for dialogs              | Smart card KSPs               | B  | Not covered | Only meaningful with PromptUser (IFACE-07).                                                                                                                               | L   |
| PROP-16              | <b>NCRYPT_USE_COUNT_PROPERTY</b>         | Key usage counter                    | Some HSMs                     | C  | Not covered | Would need a counter object on the token incremented per operation. SoftHSM2 has no atomic counter primitive.                                                             | L   |
| <b>KEY LIFECYCLE</b> |                                          |                                      |                               |    |             |                                                                                                                                                                           |     |
| LIFE-01              | <b>CreatePersistedKey</b>                | Immediate and deferred generation    | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-02              | <b>FinalizeKey</b>                       | Deferred generation trigger          | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-03              | <b>OpenKey by name</b>                   | Asymmetric and symmetric             | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-04              | <b>DeleteKey</b>                         | Destroys all token objects           | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-05              | <b>EnumKeys</b>                          | Paged enumeration with state         | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-06              | <b>EnumAlgorithms</b>                    | NCryptEnumAlgorithms support         | All providers                 | B  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-07              | <b>IsAlgSupported</b>                    | NCryptIsAlgSupported                 | All providers                 | B  | Covered     | —                                                                                                                                                                         | —   |
| LIFE-08              | <b>Machine vs user key scope</b>         | NCRYPT_■MACHINE_■KEY_■FLAG           | All providers                 | B  | Not covered | All keys are token-global. PKCS#11 has no per-user scoping within a slot; would need label prefixing plus an ACL model.                                                   | M   |
| <b>OPERATIONAL</b>   |                                          |                                      |                               |    |             |                                                                                                                                                                           |     |
| OPS-01               | <b>Hardware key protection</b>           | Tamper-resistant storage             | All commercial providers      | A  | Not covered | Fundamental: SoftHSM2 is a software token using encrypted SQLite. Only fixable by pointing the KSP at a real HSM which the PKCS#11 abstraction already allows.            | N/A |
| OPS-02               | <b>FIPS 140-2/3 validation</b>           | Certified cryptographic module       | All commercial providers      | B  | Not covered | Follows from OPS-01. Requires certified hardware and a formal validation programme.                                                                                       | N/A |
| OPS-03               | <b>Authenticode-signed binary</b>        | Commercial code-signing certificate  | All commercial providers      | A  | Not covered | Buy an OV or EV certificate and sign with signtool. No Microsoft involvement needed. See doc 10.                                                                          | S   |
| OPS-04               | <b>Multiple slots / tokens</b>           | Selecting among several tokens       | All commercial providers      | B  | Not covered | Only the first token-present slot is used. Add a SOFTHSM2_SLOT environment variable or a provider property and thread the slot ID through P11_CONTEXT.                    | M   |
| OPS-05               | <b>Key attestation</b>                   | TPM / HSM-signed key origin proof    | MS Platform Crypto; some HSMs | B  | Not covered | No PKCS#11 standard mechanism exists. Vendor-specific and unreachable through a portable PKCS#11 layer.                                                                   | L   |
| OPS-06               | <b>Load balancing / HA</b>               | Multiple HSM members                 | Thales; Entrust               | C  | Not covered | Handled inside vendor PKCS#11 libraries. Our KSP would inherit it automatically when pointed at such a library; nothing to implement.                                     | N/A |
| OPS-07               | <b>Session pool concurrency</b>          | Concurrent NCrypt calls              | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |
| OPS-08               | <b>Re-initialisation without restart</b> | Recover from a bad DLL path          | Vendor-specific               | C  | Not covered | InitOnceExecuteOnce is one-shot per process even on failure. Would need a resettable init guard; a real fix requires care around in-flight sessions.                      | M   |
| OPS-09               | <b>PIN caching / re-login</b>            | Session recovery after token logout  | All commercial providers      | C  | Partial     | C_Login tolerates CKR_■USER_■ALREADY_■LOGGED_■IN but there is no re-login path if a session drops. Add a retry that reopens and re-logs in on CKR_■USER_■NOT_■LOGGED_■IN. | M   |
| OPS-10               | <b>Logging / diagnostics</b>             | Trace output for support             | All providers                 | A  | Covered     | —                                                                                                                                                                         | —   |

## Maintaining this document

The source of truth is `softhsm_ksp/docs/feature-matrix.csv`. One row per capability. To record that a gap has been closed, change that row's *status* to Covered and clear its *gap\_solution*; to add a newly observed market feature, append a row with a fresh ID. Then regenerate:

```
python3 generate_feature_matrix.py
```

Every number on page 1 is computed from the CSV at generation time, so the summary cannot fall out of step with the table. Because the CSV is plain text, a change to coverage shows up as a one-line diff in review rather than an opaque binary change.

Actionable gaps by effort: **4 small, 17 medium, 18 large**. Rows marked N/A are excluded — they are deliberate positions (private key export is refused by design) or external blockers (CNG defines no EdDSA identifier; SoftHSM2 is a software token).

**A caution on the market columns.** An HSM datasheet is not a statement about its CNG provider. AWS CloudHSM performs AES, DES3 and HMAC in hardware, yet its KSP answers `NCryptIsAlgSupported` for exactly four identifiers. Every grade-C cell here carries that uncertainty. See `docs/11-market-comparison.md` for the full audit and its sources.